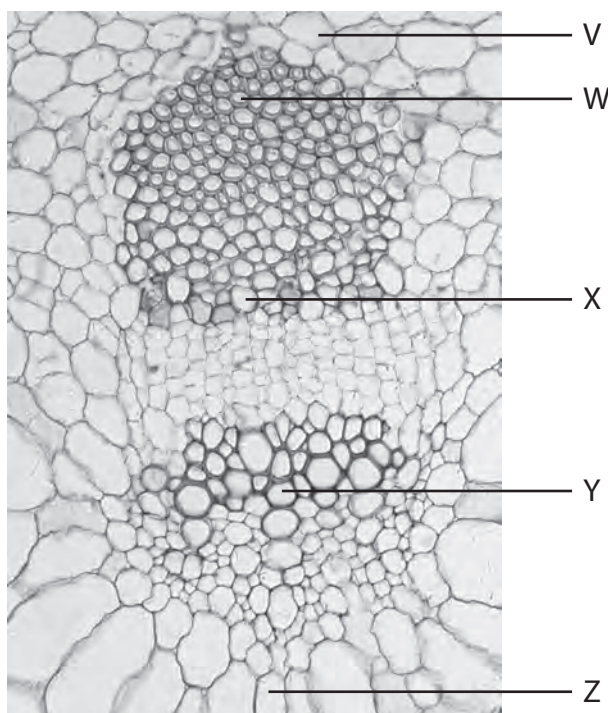


- 5 The stem of a plant contains structures involved in transporting substances and supporting the plant.

The photograph shows a section through part of the stem of a plant, as seen using a light microscope.



(Source: © Choksawatdikorn/Shutterstock)

- (a) (i) Which of the labelled structures is sclerenchyma?

(1)

- ☐ **A** W
- ☐ **B** X
- ☐ **C** Y
- ☐ **D** Z



(ii) Which of the labelled structures is involved in transporting substances both up and down the plant stem?

(1)

- ☐ **A** W
- ☐ **B** X
- ☐ **C** Y
- ☐ **D** Z

(iii) Which of the labelled structures is involved in transporting calcium ions from the stem to the leaves?

(1)

- ☐ **A** V
- ☐ **B** W
- ☐ **C** X
- ☐ **D** Y

(iv) Which of the following may have secondary thickening in the cell walls?

(1)

- ☐ **A** phloem sieve tubes only
- ☐ **B** phloem sieve tubes and sclerenchyma fibres
- ☐ **C** xylem vessels only
- ☐ **D** xylem vessels and sclerenchyma fibres

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



(b) Plants require inorganic ions for growth.

The photograph shows a leaf with magnesium ion deficiency.



(Source: © Nigel Cattlin / Science Source / Science Photo Library)

In an investigation, groups of plants were watered with the same volume of a nutrient solution.

The solution contained all the required inorganic ions, but with different concentrations of magnesium ions.

The table shows the results of this investigation.

Concentration of magnesium ions in nutrient solution / mmol dm^{-3}	Mean dry mass of root / g	Mean dry mass of shoot / g	Mean total dry mass / g
0.01	2.38 ± 0.15	15.56 ± 1.22	17.94 ± 1.38
0.10	5.42 ± 0.43	33.36 ± 1.21	38.78 ± 1.63
0.40	7.08 ± 0.50	46.06 ± 0.59	53.14 ± 0.50



(i) Explain the importance of magnesium ions to plants.

(2)

(ii) Comment on the results of this investigation.

Use the information in the table to support your answer.

(4)

(Total for Question 5 = 10 marks)



P 6 6 1 5 4 A 0 1 7 3 2